

Estimating CO₂ Emissions at UCSF Health Using a Hybrid Sampling Approach

Yuhan Xie

June 22, 2025

1 Introduction

Understanding and quantifying the carbon footprint of healthcare systems is essential for driving sustainable policies and decarbonization strategies. This study aims to estimate the total CO₂ emissions generated by procurement activities at UCSF Health using a combination of Life Cycle Assessment (LCA) data with statistical sampling methods and statistical inference.

Estimating Scope 3 emissions remains a significant challenge. UCSF Health procures approximately 25,000 distinct items, making it infeasible to conduct life cycle assessments for each manually. Current approaches that rely on industry-average emissions per unit cost are insufficient, as they can lead to serious misestimations at the product level. These inaccuracies often result in the misallocation of investments and efforts intended to reduce emissions.

To overcome these limitations, we introduce a novel hybrid approach that integrates LCA data with multiple data-driven techniques, including automated item grouping using TFIDF vectorization to consolidate similar products, and post-hoc reweighting using propensity score adjustments to mimic a probability proportional to size (PPS) sampling framework.

2 Exploratory Data Analysis

We begin with a descriptive analysis of the UCSF Health procurement dataset, focusing on both the full population and the 1,200 selected items dataset.

2.1 Full Population

For the full population dataset, the expenditure data is heavily right-skewed. Most items have relatively low expenditures, but a small number of items exceed \$1 million and even \$4 million in annual spending. To better understand this skewness, we split the expenditure distribution into high and low ranges. The histogram of low-expenditure items reveals a long-tail structure (Figure 1), while the histogram of high-expenditure items confirms the presence of a few dominant items (Figure 2).

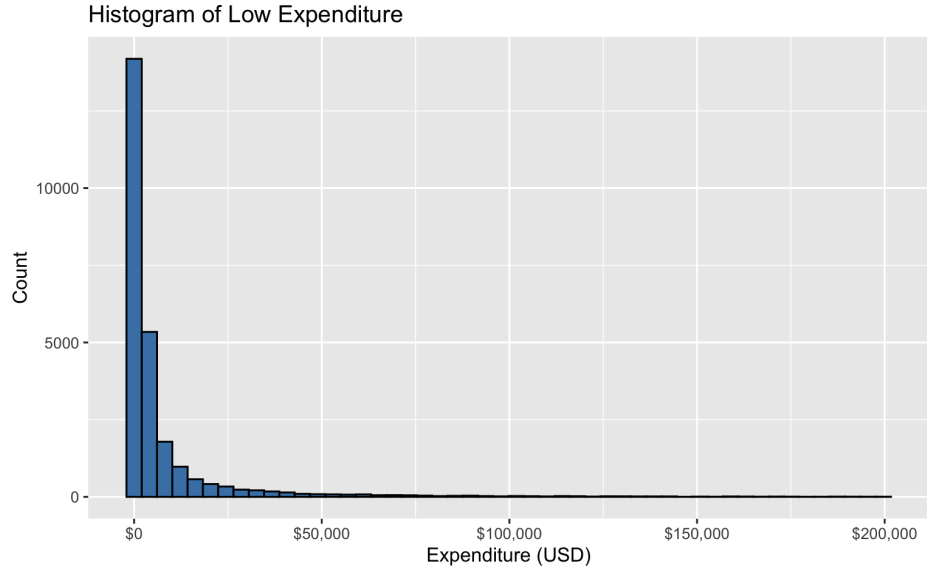


Figure 1. Histogram of low-expenditure items

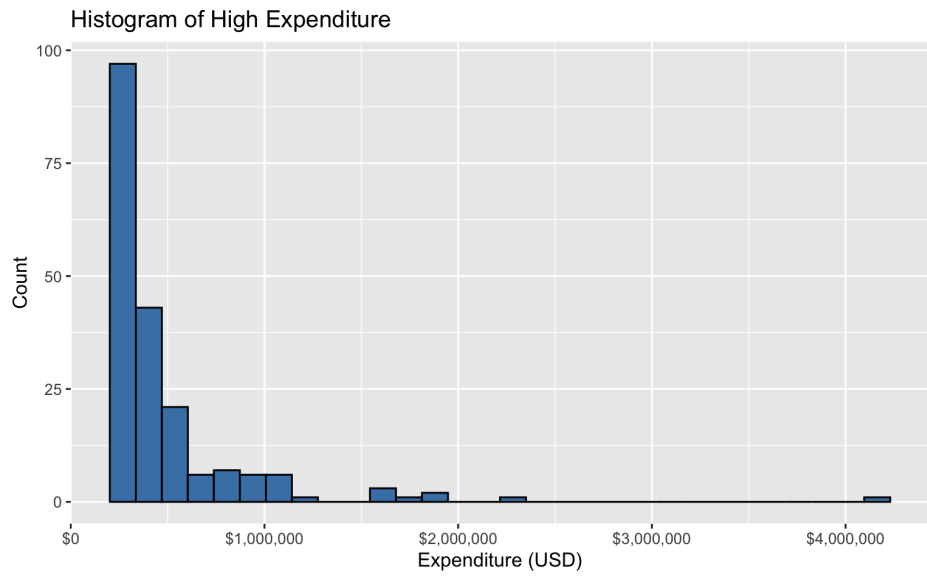


Figure 2. Histogram of high-expenditure items

Further supporting this, the cumulative expenditure plot (Figure 3) shows that a small number of items account for the majority of UCSF Health's total procurement spend. Over 80% of the total expenditure is captured by fewer than 20% of items, justifying our decision to isolate top-spending items as deterministic contributors to overall CO₂ estimates.

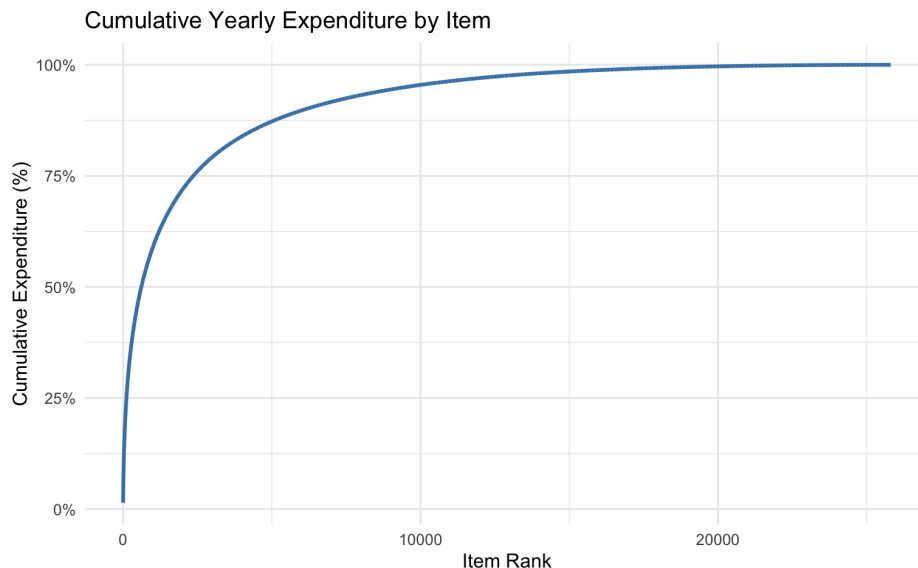


Figure 3. Cumulative yearly expenditure by item

2.2 1200 Selected Dataset

We also examine the expenditure distribution of the 1,200 products selected for sampling, and they exhibit a very similar distribution to the population. The histogram of low-expenditure items shows a gradual tapering of frequency (Figure 4), while the high-expenditure histogram shows a concentration of spending in a handful of items (Figure 5).

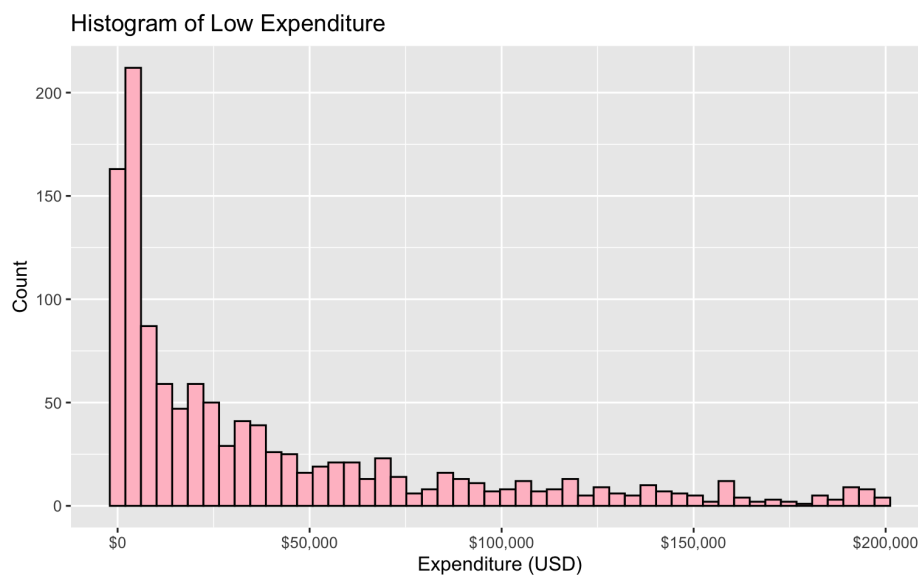


Figure 4. Histogram of low-expenditure items (selected data)

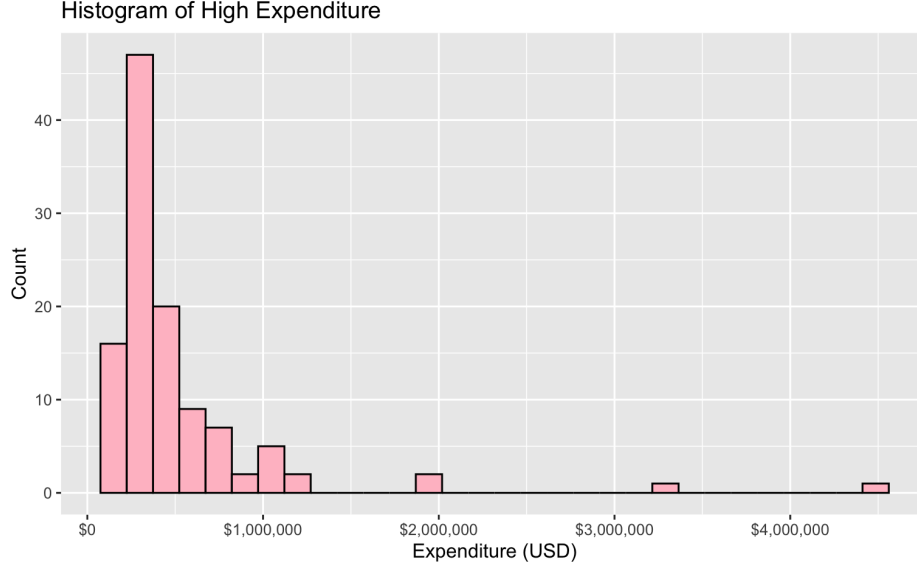


Figure 5. Histogram of high-expenditure items (selected data)

After cleaning the dataset to remove missing values and outliers, we calculated the Pearson correlation between total expenditure and total CO₂ emissions across all items. The resulting value of 0.478 indicates a moderate positive correlation. This empirical relationship provides justification for using expenditure as a proxy variable to inform sampling weights, particularly in our strategy to mimic probability proportional to size (PPS) sampling through propensity score adjustment.

3 Methodology

3.1 Grouping Items into Sampling Units

To reduce redundancy in the sampling process and improve efficiency in the life cycle assessment (LCA) workload, we group items that represent the same product but differ only in size or packaging into a single sampling unit.

- **TFIDF Vectorization:** Each item name is converted into a TF-IDF vector.
- **Cosine Similarity:** We compute pairwise cosine similarities between item name vectors.
- **Clustering Threshold:** Items with cosine similarity above 0.9 are grouped together.

ITEM_DESCRIPTION	sampling_unit_id
(CHR) CATH BALLOON NEURO 9 X 1.5MM GATEWAY M0032072209150	12284
(CHR) OCCLUDER (N/APP) MUSCULAR VSD 22MM 9-VSD-MPI-IDE-022	1209
(CHR) OCCLUDER (N/APP) MUSCULAR VSD 24MM 9-VSD-MPI-IDE-020	1209
(CHR) OCCLUDER (N/APP) MUSCULAR VSD 24MM 9-VSD-MPI-IDE-024	1209
(CHR) OCCLUDER (N/APP) MUSCULAR VSD 16MM 9-VSD-MPI-IDE-016	1209
3M RED DOT TRACE PREP (TAPE) 2236	9879

Figure 6. Example of grouping items into sampling units

3.2 Identifying Top Emitters

We manually identified the top 30 CO₂ polluters, which account for approximately 80% of total emissions in the dataset. These sampling unit items have emissions that are magnitudes larger than the rest of the products. Including them in the sample would significantly inflate the variance of our estimator.

To address this, we excluded these top emitters from the sampling and reweighting process. Their emissions are computed separately and then added as a constant to the estimate of emissions from the remaining population. This hybrid strategy reduces variance while ensuring that these dominant emitters are fully accounted for in the total emissions estimate.

3.3 Propensity Scoring and Horvitz-Thompson Estimation

Although our selected dataset of 1,200 items was not drawn through random sampling, we approximate a probability proportional to size (PPS) sample by applying propensity score reweighting. This corrects for selection bias by assigning each sampled unit a weight based on its estimated inclusion probability.

We fit a logistic regression model using total expenditure to estimate the probability $\hat{p}_i$ that sampling unit i was selected. Each item is then reweighted by the inverse of its estimated selection probability:

$$\hat{w}_i = \frac{1}{\hat{p}_i} \quad (1)$$

Using these weights, we directly estimate the total CO₂ emissions from the sampled items using the Horvitz-Thompson estimator:

$$\hat{Y}_{\text{HT}} = \sum_{i \in S} \frac{y_i}{\hat{p}_i} \quad (2)$$

We then add the emissions from the manually identified top emitters to obtain the adjusted total emissions estimate:

$$\hat{Y}_{\text{HT, adj}} = \hat{Y}_{\text{HT}} + \sum_{i \in \text{Top Emitters}} y_i \quad (3)$$

where:

- y_i is the CO₂ emissions for sampling unit i
- $\hat{p}_i$ is the estimated propensity score (inclusion probability)
- S is the set of sampled units (excluding top emitters)
- $\sum_{i \in \text{Top Emitters}} y_i$ is the total emissions from the excluded top 30 emitters

3.4 Propensity Scoring and Ratio Estimation

As an alternative to the Horvitz-Thompson total estimator, we also estimate the total emissions by first calculating a weighted mean of emissions per sampling unit, and then multiplying this mean by the number of sampling units in the population stratum (excluding the top emitters).

First, the weighted mean emission, $\bar{y}_{\text{weighted}}$, is calculated:

$$\bar{y}_{\text{weighted}} = \frac{\sum_{i \in S} \frac{y_i}{\hat{p}_i}}{\sum_{i \in S} \frac{1}{\hat{p}_i}}$$

This estimated mean emission per item is then multiplied by N_{rest} , the known total number of sampling units in the population (excluding the top emitters), to obtain the total emissions estimate for this stratum:

$$\hat{Y}_{\text{Total}} = N_{\text{rest}} \cdot \bar{y}_{\text{weighted}}$$

Finally, the emissions from the top emitters are added to obtain the adjusted final total:

$$\hat{Y}_{\text{adj}} = \hat{Y}_{\text{Total}} + \sum_{i \in \text{Top Emitters}} y_i$$

4 Results

4.1 Results for Horvitz-Thompson Estimator

We estimate total CO₂ emissions using the Horvitz-Thompson (HT) estimator, a design-unbiased method that corrects for non-random sampling through inverse-probability weighting. This approach is particularly suited for unequal probability sampling designs, such as ours, where selection probabilities are modeled using propensity scores.

- **Estimated Total CO₂ Emissions:** 39,723 tonnes
- **95% Confidence Interval:** [33,346, 46,100] tonnes

Variance for Horvitz-Thompson We compute the sample-based variance estimate for the Horvitz-Thompson total as:

$$\widehat{\text{Var}}(\hat{Y}_{\text{HT}}) = \frac{n}{n-1} \sum_{i \in S} \left(\frac{y_i}{\hat{p}_i} - \frac{\hat{Y}_{\text{HT}}}{n} \right)^2$$

Standard Error for Horvitz-Thompson The standard error is the square root of the variance:

$$\text{SE}(\hat{Y}_{\text{HT}}) = \sqrt{\widehat{\text{Var}}(\hat{Y}_{\text{HT}})}$$

4.2 Results for Ratio Estimation

- **Estimated Total CO₂ Emissions:** 42,419 tonnes
- **95% Confidence Interval:** [35,433, 49,406] tonnes

Variance for Ratio Estimation The sample-based variance for the total estimated via the weighted mean method is based on the weighted squared residuals from the estimated mean, scaled by the square of the population size. The variance of the adjusted total, $\widehat{\text{Var}}(\hat{Y}_{\text{adj}})$, is equal to the variance of the estimated component, as the emissions from top emitters are a known constant. The formula is:

$$\widehat{\text{Var}}(\hat{Y}_{\text{adj}}) = N_{\text{rest}}^2 \cdot \widehat{\text{Var}}(\bar{y}_{\text{weighted}})$$

Standard Error for Ratio Estimation The standard error is the square root of the variance:

$$\text{SE}(\hat{Y}_{\text{adj}}) = \sqrt{\widehat{\text{Var}}(\hat{Y}_{\text{adj}})}$$

Comparison The final estimates from the Horvitz-Thompson method (39,723 tonnes) and the ratio method (42,419 tonnes) are reasonably close, and their confidence intervals overlap. The primary difference between the estimators involves a trade-off: the Horvitz-Thompson estimator is formally design-unbiased, while the ratio estimator can offer greater precision, but only if there is a strong linear relationship between emissions and expenditure. Since both methods yielded similar results in this study, the final emissions estimate is shown to be robust regardless of the specific estimator chosen.

5 Discussion

The findings of this study demonstrate the successful application of a hybrid statistical approach to estimate Scope 3 CO₂ emissions from UCSF. The methodology presented here offers a scalable and efficient alternative to conducting a full Life Cycle Assessment (LCA) for every purchased item, a task that would be prohibitively resource-intensive.

A key innovation of our approach is the integration of natural language processing to group similar items, which streamlined the analysis by reducing redundancy. Furthermore, the strategic decision to isolate top-emitting products and calculate their impact deterministically was critical. As shown in our exploratory data analysis, a small fraction of items accounts for a disproportionately large share of total expenditure and, by extension, emissions. This hybrid model, which combines direct measurement for high-impact items with weighted estimation for the remainder, effectively balanced accuracy and efficiency while mitigating the risk of variance inflation.

The use of propensity score reweighting to retrospectively construct a quasi-random sample proved essential for generating unbiased estimates from the convenience sample of 1,200 items. While the moderate correlation of 0.478 between expenditure and emissions supports this choice, it also highlights a limitation. Expenditure does not perfectly predict emissions.

Incorporating additional covariates — such as product category or quantity purchased — could yield more accurate inclusion probability estimates

Another limitation is the reliance on the stated accuracy of the LCA data for the sampled items. The precision of our overall estimate is contingent on the quality of these underlying emissions data. Additionally, the cosine-similarity cutoff (0.9) for TF-IDF clustering and the selection of “top 30” emitters were chosen heuristically. A sensitivity analysis varying these parameters would help assess robustness.